PALEOMAGNETIC ISOCHRONS, UNSTEADINESS, AND UNIFORMITY OF SEDIMENTATION IN MIOCENE INTERMONTANE BASIN SEDIMENTS AT SALLA, EASTERN ANDEAN CORDILLERA, BOLIVIA[1]

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ABSTRACT

Fine-scaled stratigraphic data are combined with chronologic information provided by paleomagnetic stratigraphy to document and analyze chronostratigraphic variability of the Salla Beds, a sequence of distal floodplain and lacustrine intermontane basin sediments within the eastern Andean cordillera of Bolivia. Unsteadiness in sediment accumulation is identified using a conceptual model of episodic accumulation in which simulated accumulation histories are constructed using estimated rates for various fluvial facies and taking into account gaps in deposition and possible effects of erosion. Variability in the Salla Beds is compared to that in sequence of Miocene fluvial sediments in northern Pakistan to evaluate controls on spatial and temporal stratigraphic variability. Sedimentation in the Salla sequence was generally steady over short time scales, but over longer intervals infrequent carbonates, associated with unusually long hiatuses, served to reduce the net sediment accumulation rate. In the Chinji sequence of Pakistan, unsteady accumulation evolved into a pattern of steadier sediment accumulation as recurring cycles of unsteadiness were superimposed on the overall sequence. The controls on sediment accumulation over time in these two sequences may be similarly related to the presence and extent of atypical and infrequent events of deposition, non-deposition, or erosion that exert a significant effect on net sediment accumulation. The discrepancy of short-term episodic and long-term net stratigraphic accumulation rates in both these sequences suggests that the decrease in net sediment accumulation rate with increasing time reflects the superimposed effects of different levels of local geomorphic and extrabasinal controls.

INTRODUCTION

The precise age control made possible by magnetostratigraphic studies in terrestrial Neogene strata provides an unparalleled chronologic foundation for examining variability in the sedimentry record. Chronologies based on magnetic polarity stratigraphy and fission track and isotopic dating of interbedded volcanics are a powerful tool for investigating the evolution and tectonic significance of terrestrial molasse sequences in several Tertiary foreland and intermontane basins (e.g., Johnson et al. 1985, 1986; Jordan et al. 1988, 1990; Burbank and Raynolds 1988; Beer 1990). Globally correlated magnetic polarity zonations provide successions of chronologic data within the local stratigraphy that delineate relationships between tectonic events and the sedimentary responses they produce.

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Fluvial stratigraphic sequences have been conventionally viewed as an extreme case of discontinuous sedimentation. Any given fluvial stratigraphic record is the end product of many infrequent episodes of sedimentation separated by intervals of erosion and nondeposition. Successions of fluvial strata are generally regarded as "incomplete" on time scales of $<10^5$ yr (Sadler 1981; Dingus and Sadler 1982; Dingus 1982), casting serious doubts on the utility of fluvial sequences as time recorders and how meaningful studies dependent on estimates of time represented in these sequences can be.

Fluvial systems are characterized by patterns of sediment accumulation that reflect the interplay of external and internal controls: the frequency and duration of individual episodes are dependent on local conditions within the depositional basin as well as extrabasinal, allocyclic factors such as uplift rates, basin subsidence, sediment supply, and climate. These controls determine variability in rates and styles of sedimentation both laterally throughout a depositional system (non-uniformity) and over time (unsteadiness). Unsteadiness may signal changes in the dominant controls of net sediment accumulation; e.g., from the short-term influence of surficial processes to long-term

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domination by larger-scale geomorphic and extrabasinal controls. Over longer time intervals these controls interact to produce the pattern of sediment accumulation in a particular depositional system.

Using paleomagnetic isochrons, this investigation, and the companion study (McRae 1990), present two examples that identify chronostratigraphic variability in fluvial stratigraphic sequences at various levels of resolution. The analysis of high-resolution sedimentologic and chronologic data from the Lower Chinji Formation on the Potwar Plateau of northern Pakistan (McRae 1990) documents patterns of unsteadiness and nonuniformity of sediment accumulation in the floodplain environment of a large, throughgoing river system in the foreland basin along the southern margin of the Himalayan collisional orogen. The present paper (1) describes the physical stratigraphy and associated sedimentiology of the Salla Beds, a sequence of distal floodplain and lacustrine intermontane basin sediments situated within the eastern Andean Cordillera of Bolivia; (2) documents and analyzes, in a manner similar to that in the companion study, the chronostratigraphic variability of Salla using fine-scaled stratigraphic and paleomagnetic data; and (3) summarizes and synthesizes the results from both studies and evaluates controls on stratigraphic accumulation in terrestrial basins.

PHYSICAL STRATIGRAPHY AND SEDIMENTOLOGY

Regional Geologic Framework.—The Salla-Luribay Basin is situated within the eastern Andean Cordillera, near the eastern margin of the Altiplano, a major intermontane plateau (elevation above 3800 m) that extends north-south from Peru to Argentina. The basin is bounded by deformed and locally metamorphosed Paleozoic sediments (Martinez and Tomasi 1978; Lavenu 1986) and is one of several small intermontane basins on the margins of the Altiplano where significant thicknesses of non-marine Tertiary sediments are exposed. The Salla deposits are the site of a diverse suite of extinct fossil vertebrates that have been extensively collected by Bolivian, French, and North American paleontologists since the mid 1960s. Most previous investigations were directed toward paleontological description (Marshall et al. 1983; Patterson and Wood 1982; Wolff 1985; MacFadden 1985); the geology has only generally been described (Villaroel and Marshall 1982; MacFadden et al. 1985).

The basin sequence consists of two main stratigraphic units, the Luribay conglomerate, resting unconformably on Paleozoic rocks, and the Estratos Salla, or Salla Beds, a sequence over 500 m thick of variegated siltstones, mudstones, and minor sandstones which rest in gradational to slightly discordant contact upon the Luribay strata. The Salla Beds have been subdivided into a number of informal units based on general lithology, color, and characteristic fossil occurrences (MacFadden 1985, pers. comm.).

The Salla Beds were originally postulated to be of early Oligocene age, based on a similar level of mammalian faunal evolution to the type Deseadan (early Oligocene) land mammal age dated in Patagonia (Marshall et al. 1977). The paleomagnetic study of MacFadden et al (1985) established a polarity zonation for an extensive portion of the Salla sequence that enabled subsequent age correlations between several key biostratigraphic and lithologic horizons and the global geomagnetic time scale. K/Ar and fission track dates (MacFadden et al. 1985; Naeser et al. 1987) on biotites and zircons from interbedded volcanic tuffs within the Salla sequence, combined with the distinctive magnetic polarity zonation, strongly suggested a correlation to the MPTS in the lower portion of chron 6 (28–24 Ma). In addition, cross-cutting relationships between K/Ar dated igneous batholiths and volcanics on the Altiplano and the neighboring Cordillera Réal and basin-fill sediments on the Altiplano similar to those at Salla suggest that basin development may have been completed by 19–26 my (Evernden 1977).

Stratigraphic Field Studies.—The badlands topography and almost complete lack of vegetation due to the high elevation and arid climate result in near complete exposure of the entire Salla sequence and provide an ideal setting for studies of lateral and vertical stratigraphic variability. Three stratigraphic sections originally described by MacFadden et al. (1985) were re-described, measured, and sampled in detail. These three sections: Pasto Grande, El Planimiento, and Calabozo

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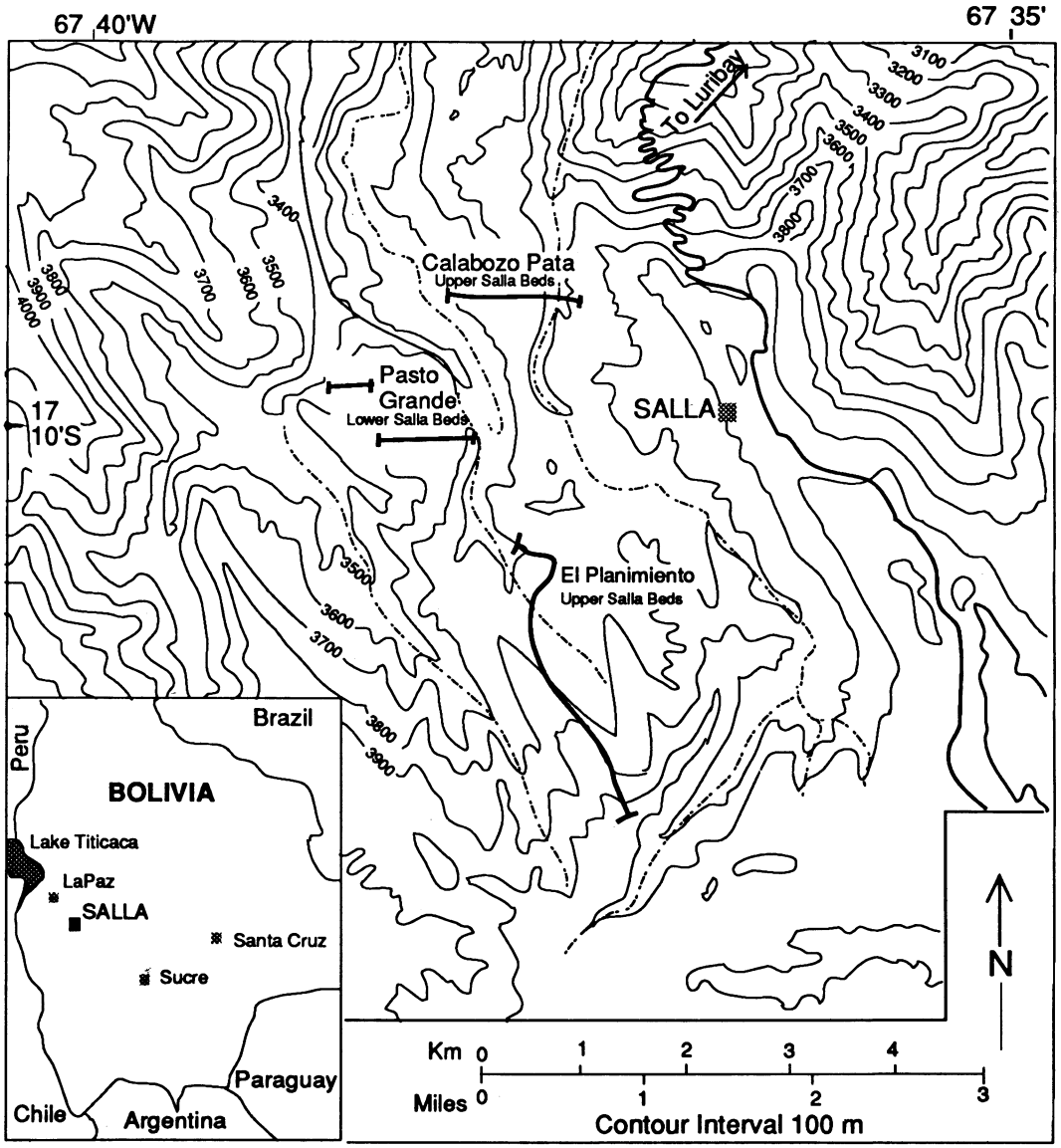


FIG. 1.—Topographic map of the Salla-Luribay basin showing locations of stratigraphic and paleomagnetic sections.

Pata together span the entire Salla sequence; the El Planimiento (EP) and Calabozo Pata (CP) sections span the same stratigraphic interval and form the basis of the detailed lateral chronostratigraphic variability study emphasized in this investigation (fig. 1). This stratigraphic interval occurs from the base of a significant volcanic unit, the Principal Guide Level of MacFadden et al. (1985) upward to another prominent volcanic unit, the Upper White Zone (MacFadden et al. 1985). These units are pervasive and have been used as the main lithostratigraphic markers for local and basin-wide correlations. The study interval corresponds roughly to the lower portion of the Upper Salla Beds (fig. 2), and comprises approximately 60–70 m of total stratigraphic thickness. The stratigraphic distance along strike between sections is approximately 1.5 km. The sections were cor-

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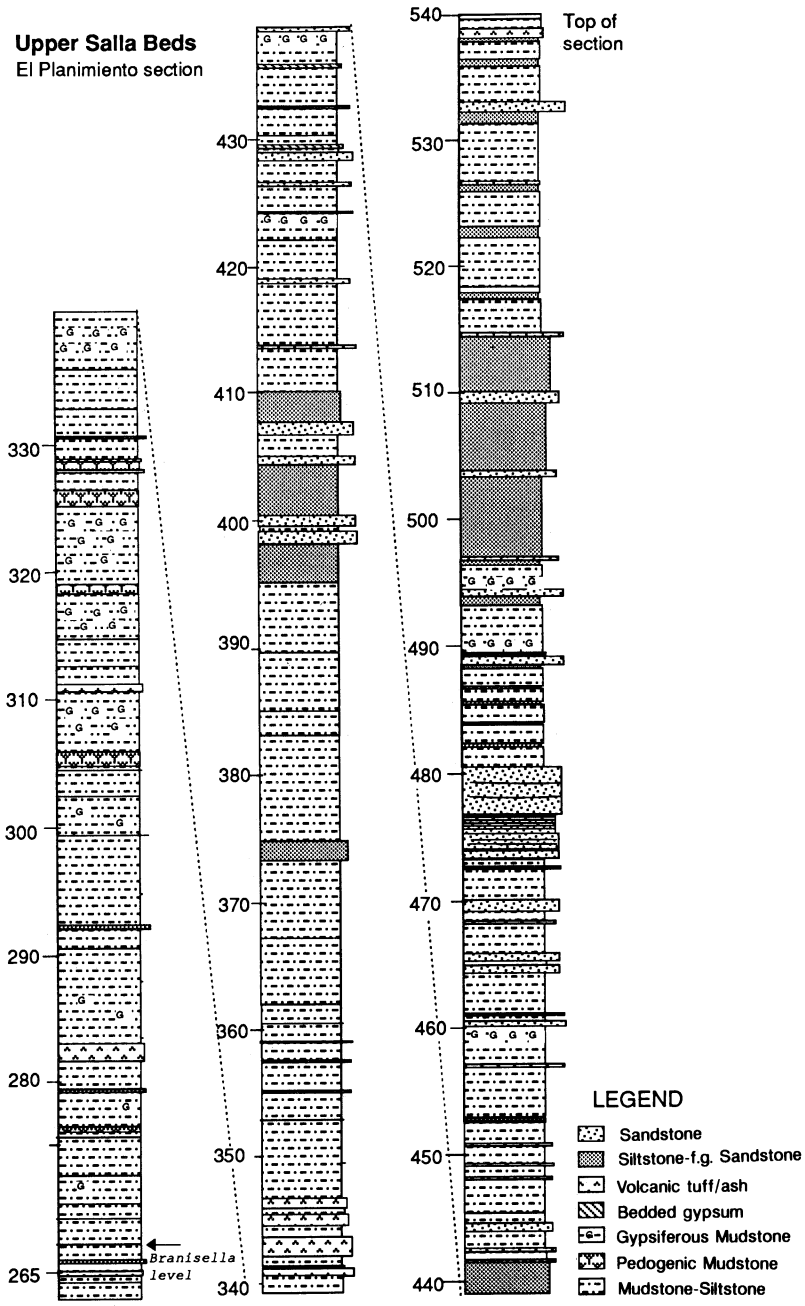


FIG. 2.—Stratigraphic section of the middle and upper Salla Beds measured along the axis of the basin (El Planimiento Section). The section that is the focus of this investigation is located stratigraphically below the sequence illustrated here and is presented in figure 3. Important biostratigraphic zones (*Branisella* level) and the location of dated volcanic horizons are indicated.

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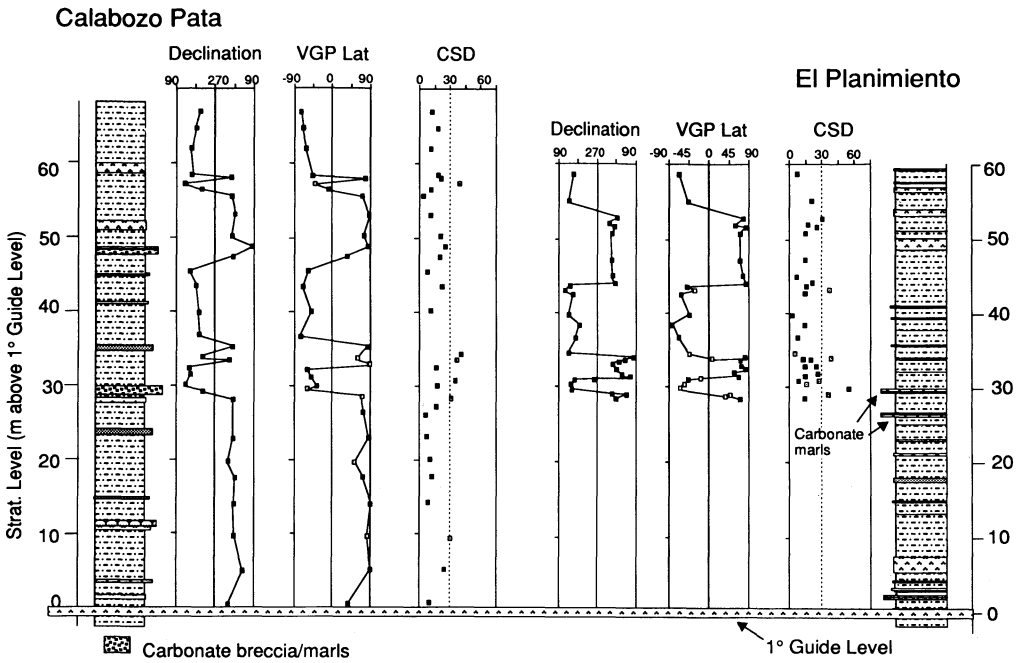


FIG. 3.—Lateral correlation of physical and magnetic stratigraphies for El Planimiento and Calabozo Pata sections. Upper contact of the Principal (1°) guide level is used as the horizontal datum for correlation between the two sections. In addition to the obvious similarities between the general pattern of magnetic zonation defined by the paleomagnetic data of successive sites, subtle directional changes in the magnetic field vector within polarity zones and near-polarity transitions are present in the same levels in both sections.

related by tracing out both the Principal Guide Level and the Upper White Zone. The difference in stratigraphic thickness between sections is attributed to the presence of additional carbonate marl units at Calabozo Pata (fig. 3).

Lithostratigraphy.—The sedimentary sequence in the Salla basin consists of approximately 600 m of variegated to massive reddish brown to pinkish tan siltstones and mudstones, volcanic ash and tuff horizons, and lesser amounts of fine-grained sandstones and channel lag conglomerates. Together these lithologies suggest deposition in distal alluvial floodplain and lacustrine environments. The vertical distribution of lithologies observed in the Salla sequence is illustrated by the El Planimiento section lithologies in figures 2 and 3. Salla lithologies can be generally subdivided into five broad facies categories.

Facies 1: Conglomerates are exposed solely in the lowest levels of the basin, and are in transitional contact with the underlying Luribay Conglomerate, previously interpreted as a proximal fan deposit by Villaroel and Marshall (1982). Salla conglomerates are flat-bedded, crudely stratified, and clast-supported. The clasts nearest the base of the sequence range from boulder to cobble size, with a maximum diameter of 35 cm and a mean diameter of 15–20 cm. Maximum and mean clast sizes gradually decrease upsection to small cobble and pebble grade in size. The clasts are generally subrounded to subangular, and are composed primarily of white to pink quartzite and green sandstone. Individual conglomerate sheets or units average <1 m in thickness, are tabular in shape, and commonly fine upward (boulder–cobble–pebble). The tops of many of the units are covered with a thin sandstone unit, suggestive of late-stage, waning flood deposition. The bases of most of the conglomerates are highly irregular, indicating erosion into underlying fine-grained sediments as well as scouring previously deposited conglomerate units.

Facies 2: Sandstone. Fine-grained, yellow-

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gray to light-brown sandstones occur commonly as thin (<1 m) units intercalated with finer-grained siltstones and mudstones, and less commonly as thicker units 1–2 m thick in the lower and upper portions of the Salla sequence (fig. 2). All the sandstones observed were fine-grained and moderately well sorted. Units near the top of the Upper Salla Beds possess some degree of bioturbation; they generally have no evidence of stratification. Basal contacts between the interbedded thin sandstones and mudstone/siltstone units are usually abrupt, while their upper contacts are often diffuse. Most of the thin sandstone units are limited in their lateral extent and thin laterally within 100s of meters. The contacts between the thicker sandstone units and the underlying finer-grained units are generally abrupt, with occasional evidence of erosive scour.

Facies 3: Variegated Siltstones and Mudstones. This facies accounts for over 90% of the Salla sequence and includes grain sizes ranging from fine clay to coarse silt. Individual unit thicknesses average 1–2 m. Some of the sandy siltstone units appear horizontally laminated, but otherwise little primary bedding structure has been preserved. The majority of the fine-grained siltstones and mudstones possess evidence of various degrees of pedogenic alteration. Evidence of pedogenesis includes green-gray Fe reduction halo mottling caused by burrowing and/or root action, disseminated spherical carbonate nodules (Walker 1988), and the slickensides and blocky ped fabrics such as those commonly attributed to clay translocation during the periodic wetting and drying cycles that commonly occur during soil development (Freytet and Pleziat 1982). The red, brown, and buff color banding is the most obvious distinguishing feature of this facies; the superposition of subtly varied lithologies with their associated pedogenic alteration features gives rise to the variegated appearance of the Salla Beds. As has been the case in previous studies of Cenozoic continental sediments, there appears to be a strong relationship between the degree of pedogenic alteration and color. Multiple pedogenic features that would represent the most profound pedogenesis in the Salla rocks occur almost exclusively in dark red claystones to silty claystones. This observation is consistent with descriptions of other Tertiary fluvial sequences (Bown and Kraus 1981; Kraus and Bown 1987).

While pedogenic features are observed throughout the fine-grained sediments in the Salla sequence, units possessing evidence of severe pedogenic alteration are uncommon. This probably is a reflection of the steadiness of accumulation and the relatively rapid sedimentation rate of the basin, which restricted the time of non-aggradation necessary to develop deep soil profiles. For this reason, the Salla units classified as paleosols, for the purposes of modeling sediment accumulation histories presented later, will be limited to those beds possessing evidence of intense pedogenic modification.

Facies 4: Volcaniclastic Sediments. Numerous volcanogenic horizons, which represent both primary air fall and secondarily reworked ash and tuff deposits, are present throughout the Salla sequence. These units are laterally continuous, and many have been used for stratigraphic correlation (MacFadden et al. 1985). Most units are fine-grained volcanic ashes. The ashes that occur either as single, relatively pure massive units or as thin (< 0.5 m) beds intercalated within mudstone units of equivalent thicknesses appear to have been deposited from primary airfalls that accumulated in standing water either on a floodplain or within a shallow lake. Secondary reworked tuffs are also present within coarser-grained, sandy siltstone units. This category includes the Principal Guide Level, which consists of a complex of multiple ash and lithic tuff units interbedded with silty mudstones and sandy tuffaceous horizons that represent a succession of primary airfall events. Evidence of bedding and cross stratification in a sandy crystal lithic tuff unit near the base of the Principal Guide Level suggests that some secondary reworking occurred. The siltstone/mudstone units, which are commonly bentonitic and contain visible pyroclastic and vitric components, indicate significant volcanogenic input to the Salla basin.

Facies 5: Limestone-Calcarenite Breccias. Distinctive limestone-calcarenite breccia and marl-like units present in the middle of the Salla sequence are light gray to light pink in color and possess a variety of internal fabrics. These units form thin resistant ridges that distinguish them from the more easily

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weathered clastic sediments. A prominent carbonate couplet, collectively referred to here as the "Planimiento Marl," can be traced throughout much of the basin and has provided a reliable guide horizon for recording and correlating fossil localities (Villaroel and Marshall 1982; MacFadden et al. 1985). Examination of the vertical and lateral stratigraphic relationships of these marl units from El Planmiento to Calabozo Pata indicate both lithologic and textural variations. The couplet from the El Planimiento section contains units that more closely resemble a massive limestone and a brecciated sandy carbonate unit. The Calaboza Pata section contains two calcareous breccia horizons at approximately the same stratigraphic level as the two units at El Planimiento. Each of the brecciated units rests on an eroded surface with variable surface relief. The texture and occurrence of these units on irregular surfaces suggest that they are not primary lacustrine carbonates but represent secondary accumulations of reworked and redeposited carbonate material. The origin of these units has been interpreted as near-surface pedogenic or groundwater calcretes locally reworked and swept into channels (P. F. Friend pers. comm. 1987). This explanation would support the observed variable texture and thickness of these units throughout the basin. Although the original thickness of the near-surface carbonate probably was relatively uniform, the subsequent reworking and local transport would result in a variable thickness of secondary accumulation upon redeposition. This scenario also would necessarily involve a significant period of time—a substantial period of non-deposition for the carbonate unit to develop and additional time during which the carbonate was subaerial and actively being weathered. Given these conditions, it would be reasonable to assume these units represent a significant hiatus relative to the other fine-grained units in the Salla sequence.

MAGNETIC POLARITY STRATIGRAPHY OF THE SALLA BEDS

Previous Investigations.—A composite magnetic polarity zonation for the Salla sequence was previously established by MacFadden et al. (1985) from three measured sections that collectively span the entire Salla Beds. The mean vertical separation between samples in their study averaged approximately 6 m. The superposition and pattern of these reversals contain 14 superposed magnetozones. Most of the reversals in the sequence are concentrated in the lower two-thirds of the section; the upper third is characterized by a period of almost continuous reversed polarity. For this reason, the interval targeted for additional detailed sampling for this study was confined to the magnetically "busiest" portion of the sequence encompassing the five magnetic polarity zones stratigraphically above the Principal Guide Level. Detailed sampling was performed in order to identify more accurately the stratigraphic level of reversals, and thus define more accurately the stratigraphic thicknesses contained between two reversal isochrons.

Field and Laboratory Methods.—Triplicate block samples were taken exclusively from unweathered siltstones and mudstones according to the method first described by Johnson et al. (1975). Initial samples were taken from sites with a mean spacing of 3.3 m in the stratigraphic interval from below the Principal Guide Level to above the Upper White Zone (figs. 3 and 4). Subsequent sampling in the vicinity of paleomagnetic reversals in the two sections was accomplished at 0.5 m intervals. All measurements were performed on a Schonstedt SSM-1 spinner magnetometer. Based on the consistent results from alternating field and thermal demagnetization studies from this investigation and MacFadden et al. (1985), thermal demagnetization was determined to be the more effective magnetic cleaning treatment in removing potential secondary magnetic components that may have been acquired from the present day field, past fields, or low temperature oxidation reactions. Typical behavior of Salla samples demonstrated that, after removing a very low blocking temperature component in the region of 100°–200°C, a higher temperature component was isolated from 300°–600°C. In this investigation, to clearly establish the subtleties of paleomagnetic behavior of the various Salla lithologies, one sample from each site was subjected to stepwise thermal demagnetization in 100°C steps from 200°–500°C, and then in closer temperature steps until the NRM intensity of the sample

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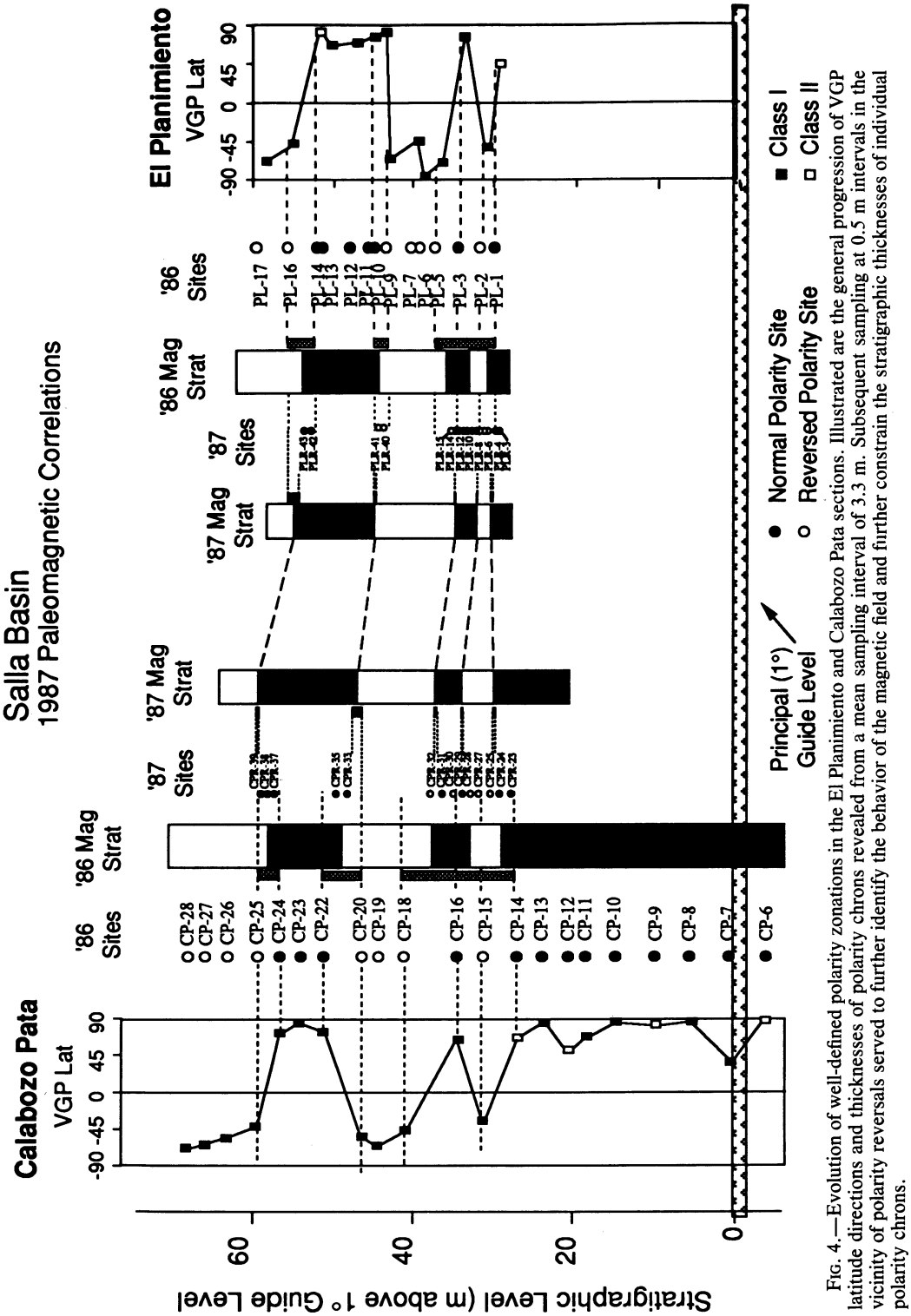


FIG. 4.—Evolution of well-defined polarity zonations in the El Planimiento and Calabozo Pata sections. Illustrated are the general progression of VGP latitude directions and thicknesses of polarity chrons revealed from a mean sampling interval of 3.3 m. Subsequent sampling at 0.5 m intervals in the vicinity of polarity reversals served to further identify the behavior of the magnetic field and further constrain the stratigraphic thicknesses of individual polarity chrons.

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had decreased sufficiently to rule out the presence of any additional, high blocking temperature component or until a stable higher temperature component was isolated. The remaining samples for each site were subjected to a blanket treatment of 400°C, and then demagnetized in progressively higher temperature steps until a stable primary magnetic component was isolated. Details of the demagnetization procedures and results are presented by McRae (1989). Based on the persistence of a higher temperature component in excess of 600°C, and its resistance to AF demagnetization, the characteristic, primary magnetic component in the Salla rocks is interpreted to be carried by hematite (MacFadden et al. 1985; McRae 1989).

Paleomagnetic Results and Data Analysis.—After thermal demagnetization studies, standard spherical statistics (Fisher 1953) were applied to test the significance of the Salla paleomagnetic data collected in this investigation. The mean directional data from statistically reliable sites clearly separate into two populations, one on the northern hemisphere possessing negative inclinations (normal polarity) (Declination = 357.2, Inclination = −24.9, N = 42, R = 39.2, K = 14.9, α_{95} = 5.9) and the other in the southern hemisphere with positive inclinations (reversed polarity) (Declination = 169.4, Inclination = 45.0, N = 30, R = 27.2, K = 10.5, α_{95} = 8.5). These clear separations are additional evidence that the primary, characteristic components of magnetization were effectively isolated. The two populations of normal and reversed samples pass a reversal test (McElhinny 1973) at the 95% condidence level (McRae 1989).

The site locations, lithostratigraphy, and details of the individual polarity transitions for the El Planimiento and Calabozo Pata sections are presented in figure 3. Directions from individual sites have been converted to their corresponding VGP Latitudes, which are plotted along with declination and the circular standard deviation (CSD), a statistical measure of the consistency of magnetization for an individual site. The similarities in the pattern of polarity zonations are obvious. There is evidence in both sections of transitional directions in VGP in the vicinity of polarity reversals, as well as evidence of increased incoherency of magnetization, indicated by a subtle "spike" of higher CSD values around the vicinity of the field transition. This may possibly be an artifact of poor directional site coherency due to incomplete demagnetization, or it may actually reflect the result of a weaker, less stable remanence acquired in the weaker geomagnetic field that persisted during the polarity transition. This phenomena has also been observed in Miocene Siwalik Group sediments of Pakistan (Tauxe and Badgley 1988; McRae 1989).

Correlation of Magnetic Stratigraphy of the Salla Sequence to the MPTS.—The stratigraphic distribution and correlation of magnetozones in the Calabozo Pata and El Planimiento sections are presented in figure 4. Detailed resampling define well-constrained magnetic polarity zonations, which were physically correlated by tracing the upper contact of the Principal Guide Level. Figure 4 illustrates the variability in stratigraphic thicknesses between the two sections, as the same chron intervals are represented by varying thicknesses of sediment. In figure 5, the detailed local magnetic stratigraphies developed at Calabozo Pata and El Planimiento are correlated to the Salla magnetic stratigraphic sections of MacFadden et al. (1985). The composite magnetic polarity zonation of the entire section was correlated to the MPTS by MacFadden et al. (1985) using the distinctive sequence and pattern of polarity reversals as well as considering the stage of evolution of the mammalian fossils and the K/Ar date of 26.5 ± 1.0 Ma from a tuffaceous horizon located at the base of the Principal Guide Level. The distinctive long period of reversed polarity was used as the principal anchor point for the MPTS correlation and was correlated to the zone of reversed polarity located in the lower two-thirds of Chron 6, indicating that the Salla Beds span 28.5 to 24 Ma (MacFadden et al. 1985). Additional results from fission track dates (Naeser et al. 1987) also suggest an age for the Salla sequence within the 20–30 Ma range. By anchoring the correlation on several fission track dates on zircons from tuffaceous horizons throughout the sequence and including new polarity zones in the lower Salla Beds identified in this study, the se-

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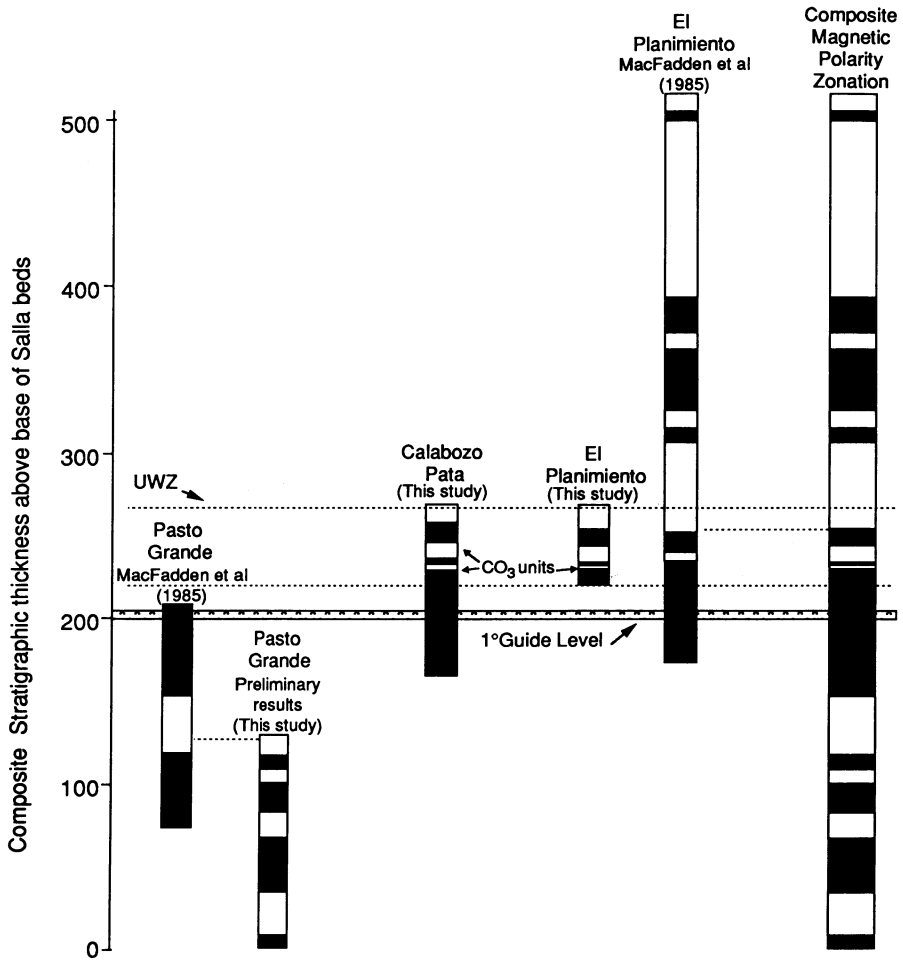


FIG. 5.—Correlation of magnetic polarity stratigraphies developed for the Salla Beds from this study and by MacFadden et al. (1985) and the composite magnetic polarity zonation.

quence may also be correlated entirely within Chron 6. This would imply a slightly younger age range for the Salla Beds, between 24–20 Ma. Neither correlation is unique; it is clear that additional paleomagnetic sampling and further resolution of the isotopic and fission-track ages are necessary to establish an unambiguous age correlation (fig. 6). For purposes of identifying and measuring chronostratigraphic variability, however, the choice of one specific correlation is not of great importance here, as each the interval spans approximately the same amount of time, the only difference being that the one suggested in this study is 4 Ma younger than the earlier correlation presented by MacFadden et al. (1985).

IDENTIFICATION AND ANALYSIS OF CHRONOSTRATIGRAPHIC VARIABILITY OF THE SALLA BEDS

Methodology.—Hypothetical sediment accumulation histories for the Salla sequence can be developed by combining the stratigraphic data from measured and described vertical sections with the time constraints provided by globally correlated magnetic reversal timelines throughout the sequence. The theory and methodology behind this approach was first introduced by Friend et al. (1989), and the specifics of the conceptual model of sedimentation accumulation in fluvial sequences and its application to an actual rock sequence are presented in a companion paper (McRae 1990); therefore, only

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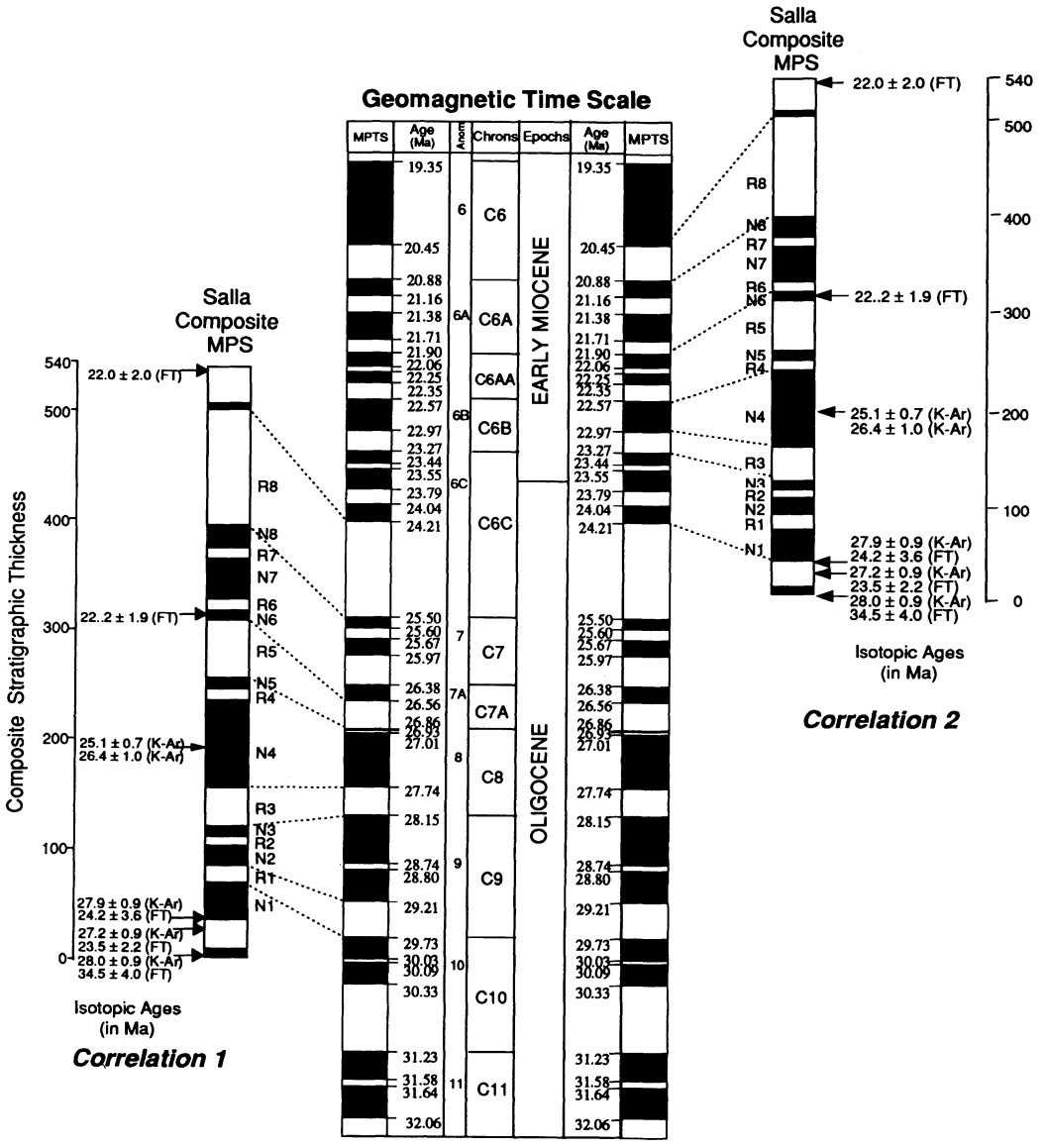


FIG. 6.—Correlation of magnetic polarity zonation of the Salla Beds to the global geomagnetic polarity time scale of Berggren et al. (1985) as presented by MacFadden et al. (1985) and suggested by this study.

an outline of the procedure is discussed here. Simulation of individual episodes of deposition is based on the assignment of arbitrary minimum rate estimates for various fluvial facies deposited in modern environments (i.e., 0.1 mm/yr for sandstones and conglomerates, 0.001 mm/yr for siltstones and mudstones, 0.0005 mm/yr for fine-grained units possessing evidence of severe pedogenic alteration [mature paleosols]). It is stressed that these rates are minimum estimates selected to exploit the large differences in magnitude between rates of high-energy, geologically instantaneous, sand deposition and the much lower accumulation rates associated with quiet water, suspended-load, mudstone sedimentation. These short-term rates of sediment accumulation are so high compared to net stratigraphic accumulation rates that changing these estimates by one or even two orders of magnitude would not significantly affect the analysis. Other mathematical mod-

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els would yield different quantitative measurements of unsteadiness, but likewise provide a basis of comparison among different stratigraphic sequences and over different time spans. Sediment accumulation histories generated by this analysis can provide a useful basis for assessing the impact of particular short-term surface events on the overall long-term accumulation rate, characterizing unsteadiness for a particular vertical sequence, and comparing unsteadiness in coeval stratigraphic intervals.

Boundary conditions used for simulating sediment accumulation at Salla take into account that individual units as described in the field are considered discrete episodes of sedimentation, and that contacts between these units are taken to represent equal periods of non-deposition (hiatus) that occur between individual episodes of sedimentation. Of course, it is likely that the individual lithologic units observed as distinctive units in the field contain even shorter-term events of episodic sedimentation of the same lithology, but at the scale of general observation, episodes are defined solely on the basis of discrete bedding contacts between units. It is also unlikely that gaps separating individual episodes of deposition represent equal amounts of time, but as it is impossible to identify the duration of most time gaps, the "equal time gap" assumption serves as a starting point for analysis.

Units possessing evidence of extreme pedogenic alteration are assigned a slower rate of accumulation, instead of a longer period of non-deposition, since a periodic incremental addition of sediment characterizes pedogenic processes in many fluvial floodplain settings (Kraus and Bown 1987). There is no evidence of erosion in the portion of the Salla sequence chosen for analysis at this time scale; therefore the contacts between thin silty sandstone and sandstone units are given the same time span of hiatus as contacts between fine-grained units. Evidence of extreme periods of non-deposition is associated with the reworked carbonate breccias present in the lower three reversal transition intervals. These units are assigned greater weight (arbitrarily, twice the time associated with "normal" hiatuses) than the hiatuses at the contacts between the carbonate units and the overlying mudstone/siltstone units.

The chronologic framework provided by paleomagnetic isochrons provides three classes of time intervals for studies of chronostratigraphic variability: (1) the time span associated with a polarity transition; approximately on the order of 10^3 yr (Fuller 1979; Clement and Kent 1984); (2) the time span associated with individual and multiple polarity chron intervals as determined by correlations of an established sequence of magnetic polarity zones to the global MPTS (10^5 yr); and (3) the time span associated with the local magnetic polarity zonations of an entire basin sequence (on the order of 10^6 yr). The actual time associated with geomagnetic field reversal has not clearly been established, and there is strong evidence that the interval involved in polarity transition may vary for different reversals. Assigning an arbitrary time span for an individual field reversal can provide a useful basis for comparing sediment accumulation records representing coeval time intervals, but this assumption cannot be used to compare stratigraphic records from different polarity reversals. Entire basin sequences (10^6 yr) are the intervals traditionally exploited in previous magnetostratigraphic studies that calculated rates of net sediment accumulation (e.g., Johnson et al. 1982, 1985, 1986), and the net accumulation rates identified over these longer time spans often exhibit trends linked to extrabasinal controls such as tectonism or climate.

Uniformity of Sediment Accumulation Histories.—Simulated sediment accumulation histories for five intervals associated with polarity reversals for which there is detailed paleomagnetic data are shown in figure 7. The detailed stratigraphy for these intervals was presented in figure 3. For the purposes of this analysis, it is assumed that each polarity transition took 10^4 yr to occur, an assumption that allows for the comparison of laterally correlated stratigraphic intervals representing the same reversal transition. Patterns of unsteadiness in sediment accumulation in each of the polarity transition intervals are based on the deviations of the short-term accumulation rate from the long-term stratigraphic accumulation rate. Examination of the Salla Beds for the observed time span reveals that sediment accumulation was generally steady (values of U are small). The similarity in sediment accumulation rates for the

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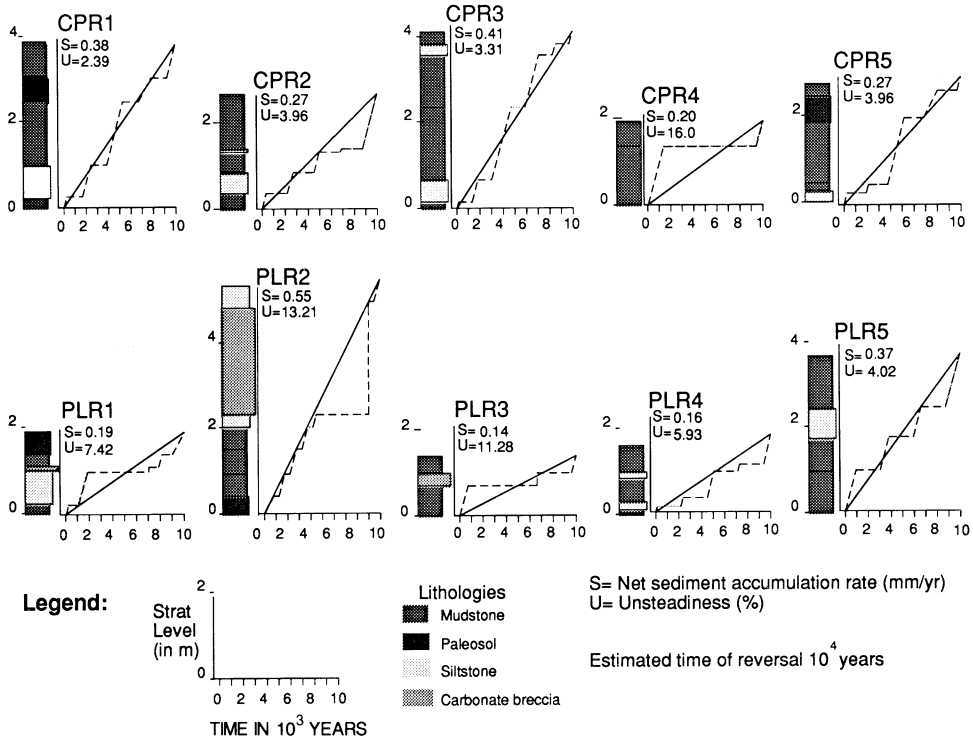


FIG. 7.—Simulated sediment accumulation histories for Salla stratigraphic intervals defined by a single polarity transition.

two coeval sequences (CPR and PLR sections) also implies that sediment accumulation was relatively uniform. The exceptions to the overall regular pattern of steady pulses of sedimentation occur where carbonate units are present. In these intervals, carbonate breccias act as a significant interruption of an otherwise regular pattern of steady sediment accumulation. At this scale of resolution, the hiatuses associated with these "rare" episodes of carbonate development are regionally important events over the alluvial surface, and the "rare" episodes subsequently dominate the net stratigraphic accumulation rate signal.

Sediment accumulation histories for intervals defined by polarity chrons provide illustrations of accumulation rate variability for time scales on the order of 10^5 yr. Figure 8 illustrates simulated sediment accumulation histories for various time spans representing single and multiple polarity chrons. The duration of a polarity zone is determined directly from the ages assigned to individual reversals on the MPTS of Berggren et al. (1985) using the second, younger correlation. The age of the Principal Guide Level is derived from an interpolation between the ages assigned at the boundary of the long normal zone surrounding the Guide Level. Comparison of the sediment accumulation histories between the two sections indicates that sediment accumulation was still relatively uniform between the two sections, i.e., as indicated by their similar rates of accumulation and unsteadiness values.

Analysis of Unsteadiness of Sediment Accumulation.—Comparison of unsteadiness of sediment accumulation (as defined by the differences in short-term episodic rates) and overall net rates of accumulation illustrate a pattern of fairly steady sedimentation, punctuated by periods of unsteadiness. These unsteady intervals appear to be the direct result of the carbonate units, and their infrequent occurrence produces a significant deviation in the general pattern of fairly continuous episodic sedimentation. It can also be ob-

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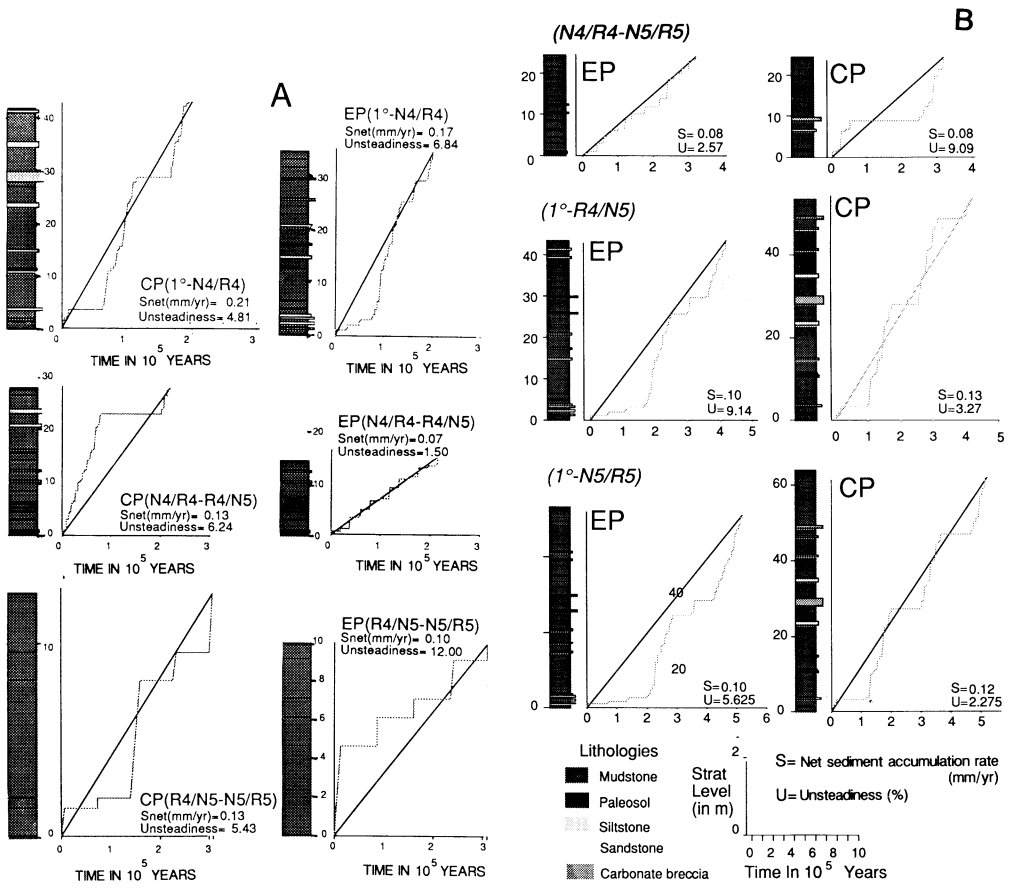


FIG. 8.—Simulated sediment accumulation histories for Salla stratigraphic intervals defined by (*A*) single polarity chrons and (*B*) multiple polarity chrons. The intervals for individual sections shown are subsets of the stratigraphic sections presented in figure 3. Section intervals are identified by the section (EP for El Planimiento, CP for Calabozo Pata) and polarity intervals (e.g., N4 to R4) that correspond to those shown in figure 6.

served that, over this longer time scale, the presence or absence of these uncommon carbonate units affects the ultimate net accumulation rate, for without these relatively long periods of non-deposition, net sediment accumulation rates would have been ultimately faster. Mean unsteadiness values of intervals on the order of 10^5 yr time spans (3.35) are slightly less than those for 10^4 yr time spans (7.15).

Examination of unsteadiness at longer time scales, on the order of 10^6 yr, is possible by considering the magnetic polarity stratigraphy of the entire Salla sequence using the age range suggested by the global MPTS. It should be pointed out that although moving to this scale necessitates using less detailed lithostratigraphic data (i.e., not at the 10^4 and 10^5 time scales), still the data are sufficient for useful analysis. The choice of a time scale does not significantly affect estimates of unsteadiness, as either potential correlation involves a time span of approximately the same duration, 4 m.y. Figure 9 presents two net sediment accumulation paths determined by plotting stratigraphic level of magnetic reversals as a function of the age of the reversal. The ages of the reversals are assigned from the MPTS correlations presented in figure 6. Although the younger correlation is presented here as more likely, the choice of one correlation over another has little effect on the pattern of net sediment accumulation over time. This is evidenced by the very simi-

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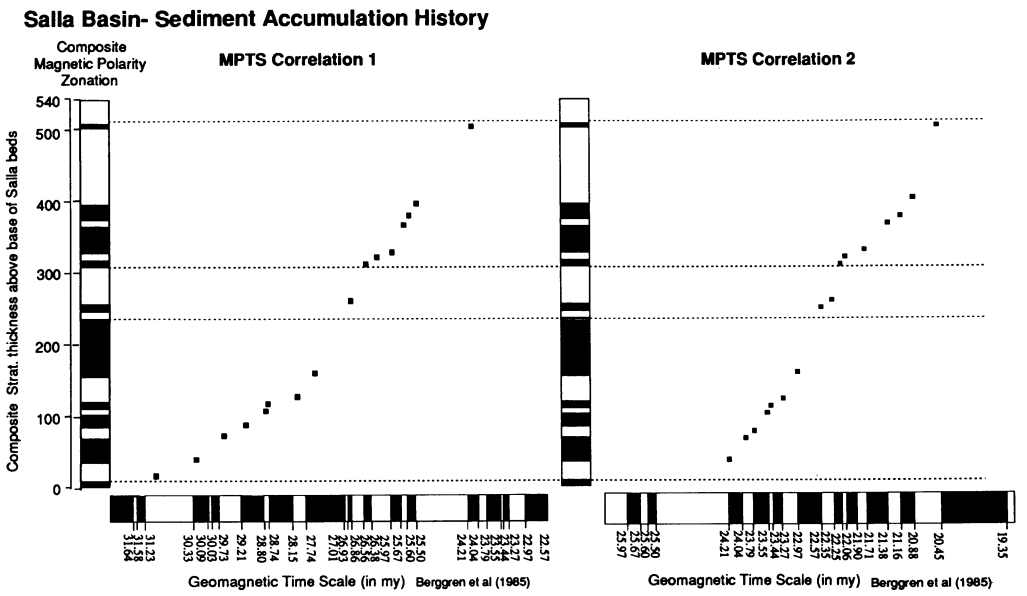


FIG. 9.—Sediment accumulation history of the Salla basin as revealed by a plot of stratigraphic level of magnetic timelines as a function of the age of the timeline. Ages of timelines are determined from the preferred MPTS correlation presented in figure 6. The slope of the line is equivalent to sediment accumulation rate. Dotted lines represent the bounding limits over which there is paleomagnetic data, the two center lines identify the interval of detailed investigation.

lar shapes of the sediment accumulation rate curves in both correlations (fig. 9).

It is instructive to compare the results of the net accumulation curve with a simulated sediment accumulation history of the entire Salla sequence (fig. 10). In generating the sediment accumulation history of the entire Salla sequence, erosion is introduced into the model because of the erosive conglomerate and sandstone units at the base, and to a lesser extent, the top, of the sequence. The similarity in the shape of the two accumulation curves, one determined from chronologic data supplied by magnetic polarity stratigraphy (fig. 10*A*) and the other from the simulated history of sediment accumulation (fig. 10*B*), supports the validity of the generalized sediment accumulation model. More important, the coincidence of the simulated sediment accumulation history with the chronologic control points provided by magnetic timelines strongly suggests that trends in sediment accumulation and changes in net sediment accumulation rates could reasonably and accurately be identified on the basis of detailed lithostratigraphy without the benefit of detailed magnetic data. The systematic deviation of the predicted rate curve from the actual data points near the top of the sequence indicates either (1) a possible miscorrelation to the time scale, or (2) that changes in the trend of sediment accumulation resulting from boundary conditions introduced into the model do not accurately reflect the actual sedimentation history of the top of the Salla sequence. The change in sediment texture and depositional style at the top of the Salla sequence, to coarser-grained more proximal clastics, as well as the problematic similarity in fission track ages separated by over 100 m of strata are evidence that the depositional history of this portion of the sequence possesses complexities not accommodated by the general assumptions built into the accumulation model.

Both curves illustrated in figure 10 also illustrate potential problems associated with previously standard methods of direct linear interpolation of sediment accumulation rates between distant data points. Part of the observed unsteadiness over this time scale (fig. 10*B*) is the result of using a single linear interpolation to define the overall net sediment accumulation rate. Inspection of sediment ac-

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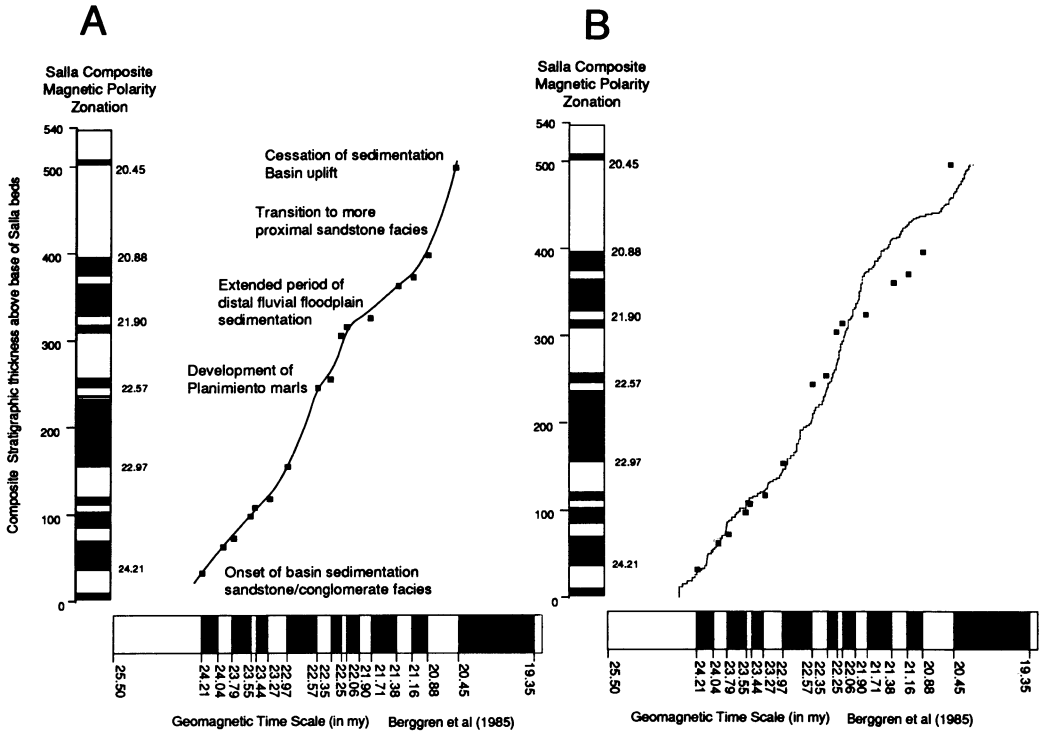


FIG. 10.—Comparisons of sediment accumulation histories of Salla-Luribay basin determined by (*A*) a plot of stratigraphic level of magnetic timelines as a function of the age of the timeline, and (*B*) a simulated sediment accumulation history for the entire Salla sequence based on the model described in the text.

cumulation rates derived from correlating magnetic time lines with the basin sequence reveals distinct breaks in slope that divide the overall curve (fig. 10*A*) into different segments. These individual segments characterize distinct phases of sedimentation at Salla, which may reflect changes in the external controls on sediment accumulation. Application of line segments of different slopes would result in generating more apparent steadiness within each of the three segments of the curve that describe sediment accumulation at Salla.

The long-term pattern of steadiness, or unsteadiness, in sediment accumulation over this time scale can be broken down into three different phases (fig. 10). The base of the sequence (0–120 m), which contains an abundance of sand and conglomerate units, is indicative of a more erosive regime. Although one can envision a smaller subset of this stratigraphic interval that may exhibit much more unsteadiness reflecting individual episodes of sedimentation, erosion, and non-deposition, over the long term the pattern is quite regular. The middle interval (120–400 m) is characterized by a notably more unsteady regime, as indicated by the greater percentage of deviation of short-term episode rate from the rate of the long-term trend. Although sedimentation was dominated by fairly regular pulses of low-energy, fine-grained, floodplain deposition, it is recorded in the overall sediment accumulation history as more unsteady. This is probably a result of the presence of "rare events" of sedimentation, the carbonate units that reflect much longer hiatuses, and that subsequently introduced a profound element of disequilibrium into the system, one which dominated the overall pattern of sediment accumulation. The latter part of the sequence (>400 m) marks a return to an interval of sandier, more erosive deposition characterized by a more unsteady pattern of deposition, erosion, and non-deposition than the middle portion of the sequence.

Trends in Non-Uniformity and Unsteadi-

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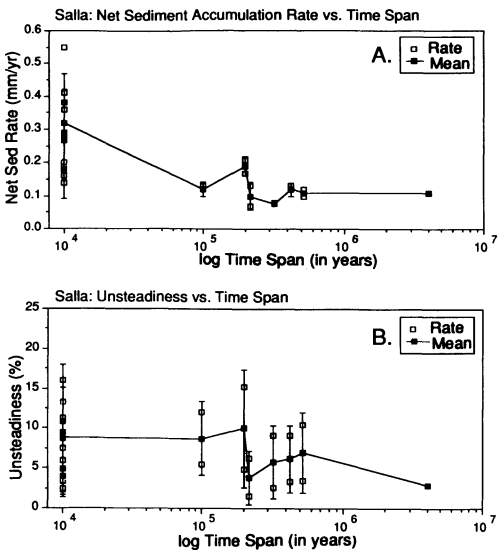


FIG. 11.—Trends in chronostratigraphic variability over different time spans for the Salla sequence: (*A*) Sediment accumulation rate vs. time span, (*B*) Unsteadiness vs. time span.

ness.—Studies of sediment accumulation histories for along strike coeval stratigraphic intervals at Salla demonstrate a considerable uniformity in sediment accumulation (figs. 7, 8). This result is not surprising, considering the lateral persistence of the majority of stratigraphic units observed throughout the basin. Examination of short time spans (10^4 and 10^5 yr) reveals subtle differences in accumulation histories for the same time interval in different sections that directly reflect the presence and extent of development of the carbonate breccia facies. The presence and thickness of these units, which have been interpreted to represent long periods of nondeposition, introduce an element of unsteadiness that directly affects the longer-term stratigraphic accumulation rates and the temporal completeness for the stratigraphic intervals in which they are contained.

Changes in net sediment accumulation rate and unsteadiness of the Salla sequence over increasingly longer time spans exhibits some fairly clear trends (fig. 11). The comparison of sedimentation rates with time span illustrates the inverse relationship documented by Sadler (1981). Spans of increasing duration also exhibit an ultimate decrease in sedimentation rate variability, with the greatest range of accumulation rates occurring over the shorter time spans (10^4 and 2×10^5 yr) and the smaller variances in rates occurring over longer time intervals. The nature of sediment accumulation at Salla can be characterized by the variation in unsteadiness vs. time span (Figure 11*B*). Over longer time spans, sediment accumulation is characterized by a generally steady pattern, but the presence of the "rare" events, the carbonate facies, continues to exercise a significant control on the longer term pattern of sediment accumulation.

COMPARISONS WITH THE CHINJI FORMATION OF PAKISTAN

This section summarizes the results from a similar study of chronostratigraphic variability of the Miocene Chinji Formation of Pakistan (McRae 1990) and compares them to the Salla sequence, in an attempt to evaluate the different factors that control stratigraphic non-uniformity and unsteadiness of sediment accumulation in these two fluvial environments and characterize the continuity of particular processes operating in these different fluvial environments.

Trends in Sediment Accumulation of the Salla and Chinji Sequences.—The Salla and Chinji sequences provide two examples of spatial variability (non-uniformity) and temporal variability (unsteadiness) in sediment accumulation in fluvial stratigraphic successions. Although there are significant differences between the two sequences, they also possess many similarities (table 1). The Salla and Chinji sequences are both Miocene in age, represent nonmarine fluvial deposition in a semi-arid climate, and were formed in regions characterized by mountain building processes that continue to the present day. The differences of the two successions lie mainly in their tectonic settings and sedimentary environments. The Salla Beds are a relatively thin sequence developed within a small restricted intermontane basin situated within the non-collisional Andean orogenic system. The Chinji sequence is part of the extensive Siwalik molasse that developed in the foreland basin in front of the large Himalayan collisional orogenic zone. The depositional environments of the two successions reflect their different tectonic settings. The Salla Beds, with the exception of alluvial fans at the base

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TABLE 1

COMPARISON OF CHRONOLOGY, TECTONIC SETTING, AND STRATIGRAPHY OF THE SALLA AND CHINJI SEQUENCES

	Salla	Chinji
Chronology		
Age:	Early Miocene (20–24 Ma)	Middle Miocene (12–14 Ma)
Stratigraphic:	500 m+	200 m+
Thickness	(complete sequence)	(lower part of 1.6 km sequence)
Net Sediment:	0.11 mm/yr	0.14 m/yr
Accum. Rates		(sequence ranges from 0.1–0.5 mm/yr)
Tectonic setting		
Source Area:	Andes	Himalaya
Basin Type:	Intermontane	Distal Foreland
Volcanism:	Frequent air falls	Rare air falls
Stratigraphy		
Lithology:	Siltstones, mudstones, sandstones, conglomerates carbonates, ashes and tuffs	Siltstones, mudstones, sheet sandstones, intraformational conglomerates
Environment:	Fluvial floodplain, lacustrine (internal drainage)	Fluvial channel and floodplain (major throughgoing river)
Climate:	Semi-arid grassland	Savanna
Fossils:	Abundant and varied	Abundant and varied

and coarser sandstones near the top of the sequence, are dominated by fine-grained, fluvial floodplain and lacustrine clastics and minor terrestrial carbonate units, facies typical of ponded intermontane fluvial systems. The Chinji sequence represents floodplain-dominated facies of a large throughgoing river system characteristic of distal foreland basin sedimentation.

Despite the obvious differences in tectonic regime and specific fluvial setting, the Salla and Chinji sequences possess similar net sediment accumulation rates (0.11 mm/yr for Salla, 0.14 mm/yr for Chinji). This similarity provides a convenient basis for comparing differences between short-term and longer-term accumulation.

In spatial variability of sediment accumulation, the Salla Beds are strikingly uniform (fig. 12). This is evidenced by the lateral persistence of facies over kilometers and similar accumulation histories, with only slight variations in net accumulation rates. The Chinji sequence demonstrates marked lateral variability in sediment accumulation over much smaller distances (100s of meters). This difference in uniformity of the two successions can be directly attributed to the different fluvial regimes. The low energy floodplain clastics deposited within the ponded intermontane basin at Salla provided for much more uniform sedimentation than did the erosive channels and overbank deposition that characterized the Chinji fluvial system. Aside from the carbonate hiatuses, accumulation at Salla was generally steady, consisting mostly of regular incremental additions of fine-grained floodplain sediment with little to no

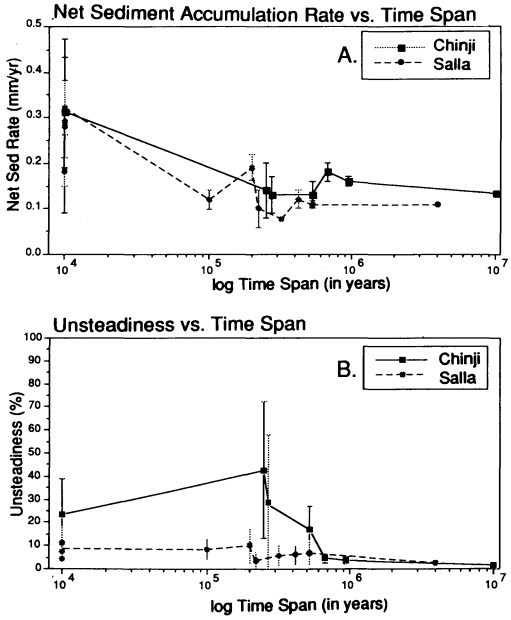


FIG. 12.—Comparison of trends in chronostratigraphic variability of the Salla and Chinji sequences: (*A*) Sediment accumulation rate vs. time span, (*B*) Unsteadiness vs. time span.

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intervening erosion between episodes of deposition (fig. 11).

At Chinji, stratigraphic intervals over short time scales (10^4 and low values of 10^5) are characterized by very unsteady sedimentation, by virtue of the varied fluvial processes operating in the system (fig. 12). Variations in sediment input from small-scale flooding events, crevasse splays, and local fluctuations in the sediment supply to particular regions of the floodplain add up to a pattern of unsteady and non-uniform sedimentation. At longer time spans ($>10^5$ yr), however, the superposition of larger-scale cycles of fluvial processes increased the abundance of "high impact" sandstone deposition events that caused unsteadiness in sediment accumulation over shorter time spans. The net accumulation rate is built up by the frequent deposition of fine-grained siltstone and mudstone, which takes the form of a background "noise" level not dominated by the signal of any one high-impact episode. The net result of this phenomenon is a longer-term pattern of sediment accumulation that appears relatively steady.

Apparent Controls of Stratigraphic Unsteadiness in Fluvial Sequences.—In fluvial systems, just what are the critical limiting processes that control temporal variability (unsteadiness) in sediment accumulation? Over short time spans (10^4 yr, which is probably the minimum level of resolution that can be accurately identified by current chronologic methods), the two case studies demonstrate two primary factors that affect a pattern of otherwise steady (i.e., not continuous, but similar magnitude, episodic deposition at regular intervals) sedimentation in fluvial environments: (1) high-energy episodes of channel sand deposition (which frequently involve erosion prior to deposition of the sand) and (2) long hiatuses (extended time of nondeposition, e.g., the development and subsequent reworking of a pedogenic or groundwater carbonate horizon). These factors were qualitatively recognized in earlier studies by Barrell (1917) and Wheeler (1956), but they did not have the advantage of fine-scaled chronostratigraphic frameworks to document them quantitatively.

Both the Salla and Chinji sequences demonstrate a subtle trend of decreasing net sediment accumulation rate with increasing time (fig. 12). This general trend has been interpreted by Sadler (1981) to result from postdepositional compaction and the likelihood of sequences to have incorporated more and longer hiatuses. The observation of these trends in the Salla and Chinji sequences suggest that the discrepancy between short-term and long-term net stratigraphic accumulation rates may be explained not by these factors, but by the superimposed effects of different levels of local geomorphic and extrabasinal controls on fluvial sedimentation. Although local controls are environmentally specific (e.g., fluvial, shallow marine, etc.), the superposition of broader extrabasinal controls (such as tectonism or climate), and their effects on any given depositional system would likely lead to a rather unique set of conditions for each basin setting.

At longer time scales, unsteadiness appears to be maintained only if the overall pattern of sediment accumulation continues to be characterized by irregular intervals of steady and unsteady deposition, nondeposition, and erosion. In the Chinji sequence, unsteady accumulation gives way to a pattern of more steady sediment accumulation as recurring cycles of unsteadiness are superimposed on the overall sequence. The inherent unsteadiness of the fluvial system has not changed, but the continued pattern of unsteadiness itself becomes steady. This phenomenon can be compared to the long-term filtering of high-frequency depositional events in sedimentary sequences proposed by Crowley (1984). Crowley presents a model of sediment accumulation dominated by low-frequency, high-magnitude events, such as the deposition of a large sand unit. "High magnitude," in Crowley's sense, describes the energy level associated with deposition. His model demonstrates that, over longer time spans, there is a tendency for high-frequency, intermediate- to low-magnitude depositional events to be "filtered-out"; that is, systematically removed from the sedimentary record. Crowley suggests that sedimentary records may not be at all representative of the depositional events that produced them because the infrequent, rare, high-magnitude events have the best potential for preservation. This scenario may be analogous to the increasing steadiness in sediment accumulation observed in the Chinji sequence. Stratigraphic intervals representing short time

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spans are commonly dominated by single, "high impact," sand units. At longer time spans, the "rare" event becomes common, and the repetitive nature of individual fluvial cycles that make up the Chinji sequences results in a high frequency of high-magnitude sand deposition events. The episodes of sand deposition are not eliminated from the stratigraphic record, as Crowley's (1984) high-frequency low-magnitude events are, but their individual impact on net sediment accumulation is lessened enough to modulate the original unsteadiness. For unsteadiness to increase over longer time spans in a stratigraphic sequence, evidence of additional, larger-scale disequilibrium processes, such as the boundaries between larger order fluvial cycles (Schumm 1977), would have to be encountered.

A similar explanation may be valid for the preserved unsteadiness of the Salla sequence. Sediment accumulation at Salla was observed to be generally steady at short time scales, except in those intervals possessing the carbonate units interpreted to represent long periods of hiatus. When the pattern of sediment accumulation is observed over longer time scales, these infrequent occurrences of carbonates, with their associated unusually long hiatuses, still impose a significant effect on net sediment accumulation rate. These are not "high-magnitude" events, in the sense of the energy responsible for their deposition (in this sense they would be of very low-magnitude). They can be considered "high impact" in the sense of the extreme hiatus that they represent. They also qualify as low-frequency events, as they are only observed at a few levels in the middle-third of the sequence. Given these considerations, these rare carbonate units maintain a significant effect on sediment accumulation, relative to the otherwise steady high-frequency, low-magnitude events associated with the bulk of Salla sedimentation.

The contrasting trends in sediment accumulation exhibited in the Salla and Chinji sequences are controlled by the time scale of sequence packaging or by depositional cycles. These time scales are ultimately related to sequence-specific autogenic (e.g., depositional setting) or allogenic (e.g., tectonics or base-level changes) processes. The actual controls on the pattern of sediment accumulation (unsteadiness) over time in these two sequences, however, may be similarily related to the presence and extent of atypical and infrequent events of deposition, non-deposition, or erosion that exert a significant effect on net sediment accumulation.

CONCLUSIONS

The chronologic information supplied by magnetic polarity stratigraphy has the potential to be exploited at several levels of time resolution; from correlations of a sequence of polarity intervals represented in an entire stratigraphic succession (10^6–10^7 yr), to tracing individual polarity reversals (10^4–10^5 yr) along strike throughout a sequence, to possibly even examining the variability associated with transitional field behavior during a single polarity reversal.

Unsteadiness in sedimentation, calculated as the mean deviation of the short term episodic accumulation rate from the long term net stratigraphic accumulation rate, is a useful measure to assess patterns in vertical variability in sediment accumulation over time and compare non-uniformity in sediment accumulation between coeval stratigraphic intervals.

Paleomagnetic evidence from the two investigations presented and compared here suggests that fluvial strata have the potential of preserving a record of sedimentation at finer time scales than previously assumed. Faithful replications of magnetic polarity reversal records in along strike stratigraphic intervals, as well as coincident observations of transitional behavior of the geomagnetic field during polarity reversal, imply that these fluvial sediments are capable recording time intervals shorter than 10^4 yr. The stratigraphic evidence of obvious and significant unsteadiness in sediment accumulation in fluvial strata over time intervals spanning a reversal (approximately 10^4 yr) implies that sediment accumulation must have been relatively "continuously episodic"; i.e., deposited at frequent enough intervals to record transitional behavior and short-term fluctuations of the geomagnetic field.

In both the Salla and Chinji sequences, the inverse trend of decreasing rates of sediment accumulation with increasing time span, previously recognized in many stratigraphic successions, has been observed. In the past,

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these general trends have been interpreted as resulting from post-depositional compaction and the likelihood for sequences to have incorporated more and longer hiatuses (Sadler 1981). The observation of these trends in the context of the two sequences studied in these investigations suggest that the discrepancy of short-term episodic and long-term net stratigraphic accumulation rates may not reflect those factors so much as the superimposed effects of various levels of internal (local geomorphic) and external (extrabasinal) controls on fluvial sedimentation.

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